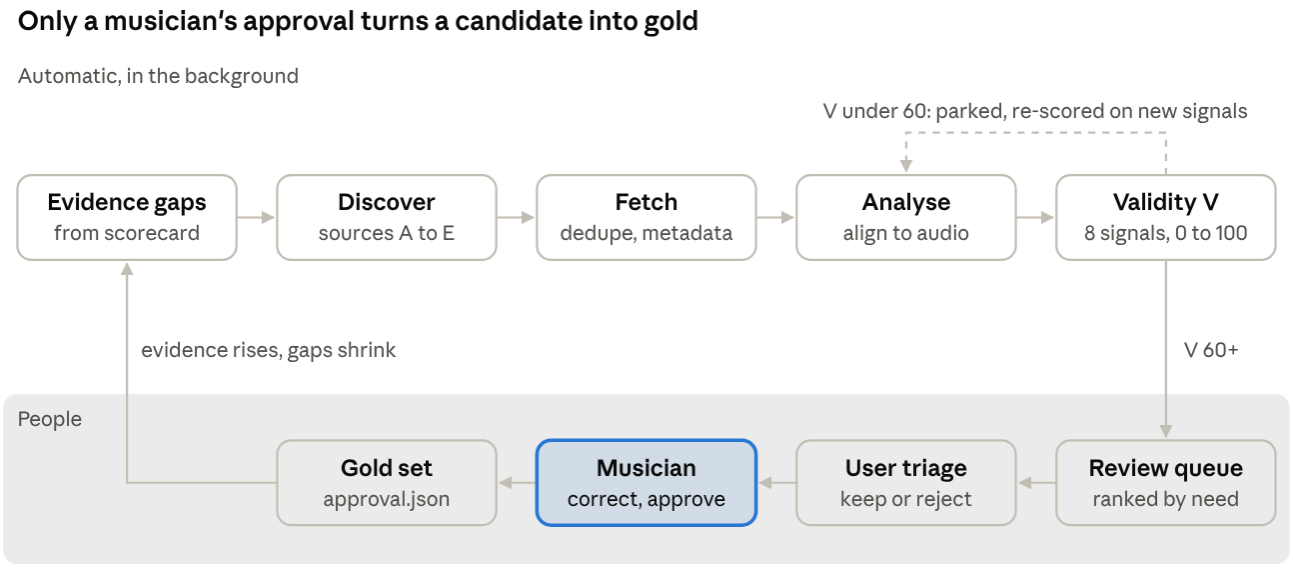
Audio Reverse Engineering — Gold Data Sourcing

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A background worker searches for music and sheet music that the scorecard is missing, checks each candidate against its audio, and gives it a validity score. The most promising candidates are surfaced for the user to triage and for an approved musician to correct and approve. Nothing becomes gold without that approval.



sourcing loop · 5 automatic stages, 3 human steps

The loop runs continuously: every approval raises evidence in the scorecard, and the next search targets whatever gap is then the largest.

What to look for

The search is driven by demand. The scorecard knows which evidence sets fall short of what is needed, and the sourcing worker fills the biggest gap first. It doesn't collect music for its own sake.

Evidence set	Need per language	EN today	LT today	Each candidate must carry
Songs with melody and word-timed lyrics	8, across ≥ 2 genres	0	3 approved	Audio + notated vocal line or MIDI + lyrics (timed if possible)

Evidence set	Need per language	EN today	LT today	Each candidate must carry
Songs with sections and chords	5	0	0	Audio + chord chart or lead sheet
Clips per instrument family (shared)	3 each for piano, guitar, bass, drums, synth, SFX	0	—	Isolated or clearly audible part + score, TAB, MIDI or hit annotations
Walkthrough songs	2	0	0	Any real song, 2–5 min

Priority order. The worker ranks open gaps by need × (1 − coverage), so whatever is furthest behind is searched first. Within a gap it aims for spread: genres, tempos, singer voice types and recording quality.

Sources

Sources come in tiers by how much we trust them before checking. Each source gets an adapter with its own rate limit, and its terms and licence are recorded at onboarding. The user carries legal responsibility, but the record still shows where every file came from.

Tier	Source	Gives us	Languages	Best for
A: curated research datasets	MAESTRO, MusicNet	Audio with aligned notes	Instrumental	Piano clips
A	GuitarSet, IDMT-SMT (guitar, bass, drums), ENST-Drums, E-GMD	Audio with note, string/fret or hit annotations	Instrumental	Guitar, bass and drum clips
A	MedleyDB, MUSDB18, Slakh2100	Multitrack stems (Slakh: rendered from MIDI, so exactly aligned)	Mostly EN	Separation, synth and SFX clips
A	DALI, JamendoLyrics	Word- or line-aligned lyrics to audio	EN (JamendoLyrics also FR, DE, ES)	Voice, EN

Tier	Source	Gives us	Languages	Best for
B: open score and lyric libraries	IMSLP, Mutopia, OpenScore	Public-domain and CC scores	Mixed; some LT	Melody and piano clips
B	LRCLIB	Time-synced lyric lines for commercial songs	EN, LT and others	Line timing for voice songs
B	MusicBrainz + AcoustID	Identity, duration, release metadata, audio fingerprints	All	Matching audio to scores and lyrics
B	Jamendo, Free Music Archive, Internet Archive	Creative Commons audio	EN, some LT	Audio for walkthrough and lyric songs
C: Lithuanian archives	Folklore and music archives, songbooks, the Lithuanian Music Information Centre (MIC.lt)	Notated melodies with lyrics, recordings	LT	Voice, LT (access terms to be checked per archive)
D: community	MuseScore community scores, Ultimate Guitar-style TAB, user-uploaded MIDI	Scores and TAB for popular songs, quality varies	All	Guitar and pop songs, only with strong agreement signals
E: our own	Commissioned scores from approved musicians; the user's own files	Reference scores and timed lyrics made to order	EN, LT	Filling gaps no source covers

Tier A and B items still need musician approval before they count fully, but they arrive as strong candidates. Tier D only enters the review queue when several signals agree (next section). For Lithuanian, tiers C and E are likely to supply most of the gold set.

Validity signals

Every candidate gets a **validity score V** from 0 to 100 before a person sees it. V estimates how likely the reference is to be correct for this audio. It ranks the review queue; it never replaces approval.

$$V = \left\lfloor 100 \cdot \frac{\sum_s w_s\, n_s}{\sum_s w_s} \right\rfloor, \quad V = 0 \text{ on any hard reject}$$

Signals that don't apply (lyrics fit for a drum clip, for example) are left out of the mean, as in the scorecard.

Signal	How it is measured	n = 1 when	Weight
Audio–score alignment	Render the score or MIDI, align it to the recording with chroma DTW; check coverage, cost, key and tempo	⩾ 95% of the recording aligned, low cost, same key and tempo	25
Cross-source agreement	Compare independent references: note F-measure between scores, chord match, lyric character edit distance	⩾ 2 independent sources agree within tolerance	20
Source tier	Prior trust in the source	Tier A (A 1.0, E 0.9, B 0.75, C 0.7, D 0.4)	15
Identity match	AcoustID fingerprint, MusicBrainz metadata, duration	Fingerprint match and duration within 2%	10
Lyrics fit	Language detected on the vocal stem; syllables vs vocal notes; timed lines vs vocal onsets	Right language, syllable count within 5%, lines within 300 ms	10
Structural sanity	Parses; ranges playable; meter consistent; TAB playable; drum map valid	No violations	10
Community signal	Ratings, downloads, "official" flags where a source has them	Top decile of its source	5

Signal	How it is measured	n = 1 when	Weight
Our pipeline agrees	Our own transcription vs the candidate	Note F ≥ 0.8	5

Hard rejects (V = 0, kept with the reason): duplicate fingerprint, unreadable file, wrong language for the gap, under 20 s, or a reference that covers less than half of the audio.

V	What happens
0–59	Parked with its reasons; re-scored when new signals arrive (a second source, better alignment)
60–84	Enters the review queue
85–100	Enters the review queue at the top, and counts as provisional (unapproved) evidence in the scorecard until reviewed

Our own pipeline's agreement carries little weight on purpose. If it counted heavily, the gold set would fill up with songs the app already gets right, and scores would look better than they are.

Background processing

A sourcing worker in `apps/remix_studio/sourcing/` runs on the workstation without blocking anything else. Each stage is resumable and records its output, so a crash or reboot loses no work.

Stage	What it does	Writes
1. Plan	Reads scorecard gaps and turns them into search tasks ("LT, voice, folk, female singer, 2 more needed")	Task list
2. Discover	Queries source adapters (APIs, dataset indexes, archive catalogues)	Candidate stubs with source, URL and metadata
3. Fetch	Downloads audio and references within per-source rate limits; checks file type and length	Files + SHA-256 under `data/remix/sourcing/<candidate>/`

Stage	What it does	Writes
4. Identify	Fingerprints the audio (AcoustID) and drops duplicates; fills metadata from MusicBrainz	Identity record
5. Analyse	Runs separation, pitch and STT on the audio; renders the reference; aligns the two	Alignment, stems, transcripts
6. Score	Computes every signal and V; applies hard rejects	`candidate.json` with signals, V and reasons
7. Queue	Puts V ≥ 60 in the review queue, ranked by gap priority × V	Queue entry
8. Promote	On approval, copies the item into the gold set with its `approval.json`; on rejection, records why	Gold item or rejection record

Running rules

- **Yields to the user.** The GPU stages pause while the user has a job running in the studio, and resume afterwards.
- **Polite fetching.** Each source has a rate limit, caches responses, and gets an adapter that respects its terms. A blocked or failing source is skipped and reported, not hammered.
- **Budgets.** Disk (proposed 50 GB under `data/remix/sourcing/`) and cloud STT spend per day are capped. The worker stops a stage when a cap is reached, and says so in the report.
- **Re-scoring.** When a new signal arrives (a second source for the same song, a better aligner), affected candidates are re-scored automatically.
- **Paths and storage.** All locations come from `studio_paths.py`; candidates and signals are also recorded in the SQLite catalog (`candidates`, `candidate_signals`).

Review surface

Candidates surface in a **Candidates** area of the studio. The user triages them, and an approved musician makes the final call. These are actions to agree on, like the DoD's user actions.

ID	Who	Can…	Done when	Priority	Status
R1	User	See a candidates inbox ranked by gap and V	Filters by language, family, source, V and stage; the top row is always the most useful next review	Must	Proposed
R2	User, musician	Open a candidate and compare it with the audio	Audio plays with the reference in its family's notation and lyrics highlighted, synced; an alignment strip marks where reference and audio disagree	Must	Proposed
R3	User, musician	See why it scored what it did	V broken down by signal with a plain reason each; source, licence and fetch date shown	Must	Proposed
R4	User	Triage quickly	Shortlist, reject with a reason, or ask for more sources, each with one key; the next candidate opens at once	Must	Proposed
R5	User	Send shortlisted items to a musician	Assign a reviewer by language and instrument; the musician sees a personal queue	Must	Proposed
R6	Musician	Correct the reference in place	Notes, chords, lyrics and timings are editable with the same editor as V11; corrections are recorded under the musician's name	Must	Proposed
R7	Musician	Approve or reject, with scope	Approval covers notes, lyrics, timing and/or chords; it writes `approval.json`; any later change to the files voids it	Must	Proposed
R8	User	Ask the system to find references for a specific song	Paste a title, URL or file; the worker searches and reports what it found within the run	Should	Proposed
R9	User	See sourcing progress	Gaps per language and family, candidates per stage, budgets used, sources failing	Must	Proposed
R10	User	Manage the approved musicians list	Add or remove a musician and set what they may approve (language, instruments)	Must	Proposed

Safeguards

- **No circular proof.** Our pipeline's agreement is worth 5 of 100 in V. At least a third of each gold set must be songs where our pipeline disagreed with the reference before review, so the set includes hard cases.
- **Held out from training.** Gold items, and candidates in review, are never used to train or tune models. Their checksums go into a held-out list that every training script checks.
- **Spread, not volume.** Each gold set has minimum spread: ⩾ 2 genres, both male and female singers for voice, and studio as well as live or home recordings. The planner won't over-fill one corner.
- **Provenance end to end.** Every gold item can be traced back to source URL, fetch date, licence field, the signals at review, and the reviewer.
- **Independent sources mean independent.** Two community uploads copied from each other count as one source. Matching file hashes or near-identical engraving collapse into a single source.
- **Legal responsibility** stays with the user, as decided. The worker still records each source's terms so the user can decide what to keep.

Changes to the scorecard

While gold data is scarce, the score mostly tracks how far the musicians have got rather than how good the engineering is. Two changes make both kinds of progress visible.

1. **Quality and evidence are shown as their own numbers.** The report shows Q (0–100), E (0–100) and the score side by side for every deliverable and language. Engineering progress shows in Q from the first run, measured on whatever items exist, approved or not. The score itself still needs evidence.
2. **What counts as unapproved evidence (U).** Only unreviewed candidates with V ⩾ 85 count at half weight. Candidates below 85, and parked ones, count for nothing. The three existing Lithuanian songs were provided and approved by musicians, so they count fully as gold.

The scoring formula itself doesn't change.

Done when

Sourcing is a deliverable in its own right (FR-16 in the PRD) and is scored like every other.

- [ ] The worker runs unattended for 7 days, and survives a reboot without losing work or refetching
- [ ] Every open gap in EN and LT has at least 3 candidates with V ⩾ 60 in the queue

- [ ] V is calibrated: at least 80% of candidates with V ⩾ 85 are approved by musicians, measured over the first 50 reviews
- [ ] R1–R7, R9 and R10 pass their walkthroughs
- [ ] Every gold item traces back to its source, signals and reviewer
- [ ] The scorecard report shows Q, E and the score separately

Open questions

- [ ] Which sources to switch on first? Some need accounts or API keys (AcoustID, Jamendo, cloud STT).
- [ ] Budgets: disk for `data/remix/sourcing/` (proposed 50 GB) and daily cloud spend.
- [ ] Who are the first approved musicians for EN and for LT, and how are they paid or credited?
- [ ] Should tier D community sources (MuseScore, TAB sites) be used at all, or only tiers A–C and E?
- [ ] Which Lithuanian archives will give access, and on what terms?